6 **(a)** An electron is travelling at a speed v in a straight line in a vacuum. It enters a uniform magnetic field of flux density 8.0×10^{-4} T which is acting into the plane of the paper. Initially, the electron is travelling at right angles to the magnetic field, as shown in Fig. 6.1.

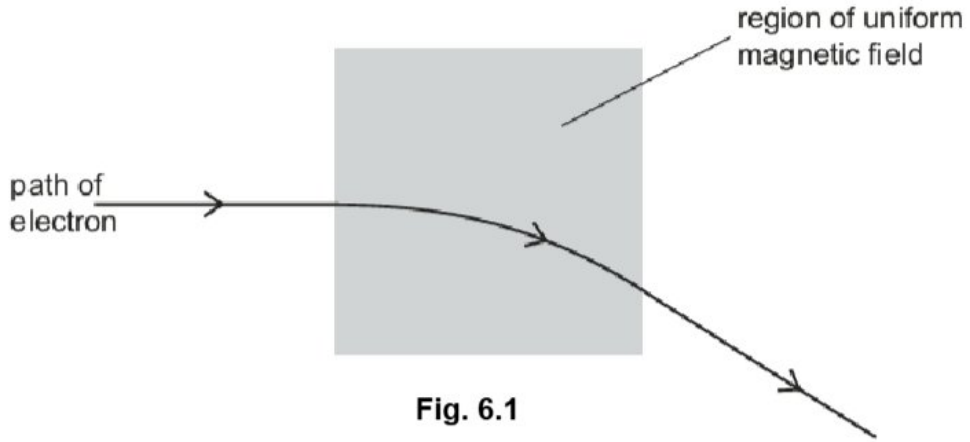


Fig. 6.1

The path of the electron in the magnetic field is an arc of a circle of radius 6.4 cm.

Determine the speed v of the electron.

speed = ... m s^{-1} [2]

(b) A uniform electric field is now applied in the same region as the magnetic field.

The electron passes undeviated through the region of the two fields, as shown in Fig. 6.2.

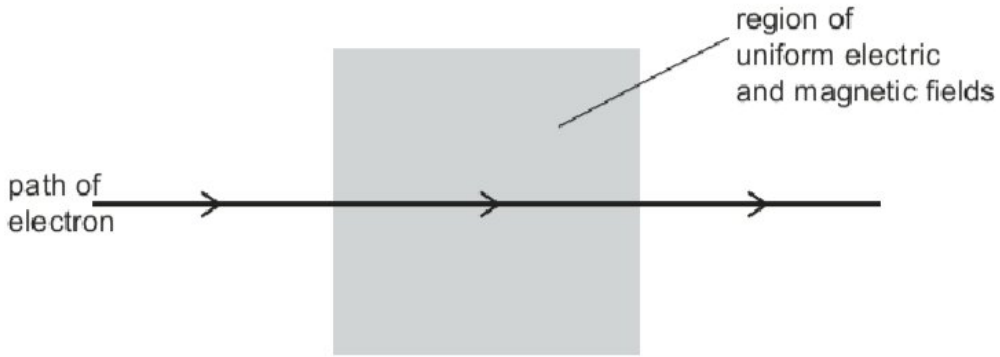


Fig. 6.2

(i) On Fig. 6.2, mark with an arrow the direction of the uniform electric field. [1]

(ii) Use the data from **(a)** to calculate the magnitude of the electric field strength.

electric field strength = ... N C^{-1} [1]

(c) The electron in **(b)** is now replaced by a positron (which has the same mass but an equal and opposite positive charge compared to an electron) travelling at the same speed *v* along the initial path as the electron.

Describe and explain the shape of the path in the region of the magnetic and electric fields.

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.....

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..... [2]

[Total: 6]